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Module 1: The Bernoulli Distribution

Story & Conceptual Intuition

Imagine standing at the door of an antique coffee shop on a rainy afternoon. A traveler steps up to the door, hesitates for a second, and makes a choice: either they push the door open and walk inside, or they turn around and walk away. There is no middle ground. They do not half-enter.

The Bernoulli distribution models this exact moment: a single, isolated trial with strictly two possible outcomes, traditionally labeled **Success (1)** or **Failure (0)**.

In everyday systems, this is a single digital light switch. It is either switched on or off.

Theoretical Foundation

Type: Discrete probability distribution

Parameter: $p \in [0, 1]$, the probability of success; $q = 1 - p$, the probability of failure

Support: $k \in \{0, 1\}$

Probability Mass Function (PMF):

$$P(X = k) = p^k(1 - p)^{1-k}$$

Expectation (Mean): $E[X] = p$

Variance: $\text{Var}(X) = p(1 - p)$

The variance reaches its highest point when $p = 0.5$ (0.25), where uncertainty is at its peak.

Business Case Study: Ad Click-Through Conversion

Scenario: An e-commerce platform displays a targeted banner for winter jackets to a unique visitor.

Interpretation: The visitor either clicks ($X = 1$) with probability $p = 0.04$ or scrolls past ($X = 0$) with probability 0.96.

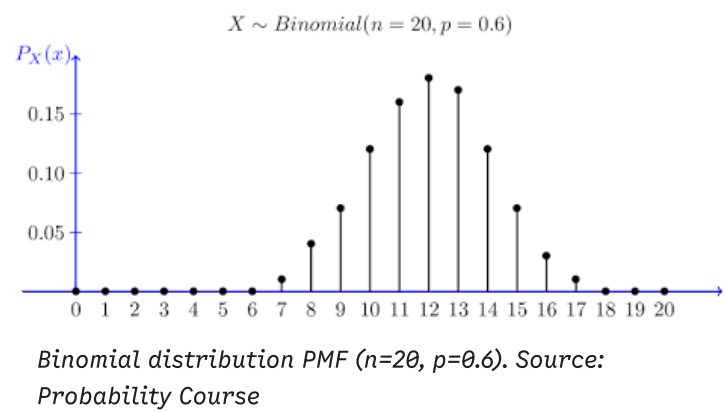
Application & Benefit: Modeling ad impressions as Bernoulli trials allows growth marketing teams to calculate baseline conversion volatility. It establishes the foundational unit used to evaluate logistic regression classifiers and run rapid A/B testing on call-to-action buttons.

Module 2: The Binomial Distribution

Story & Conceptual Intuition

Now picture our coffee shop door over the span of an hour. Exactly 20 distinct travelers approach the shop one after another. Each person makes their decision completely independently of whoever walked by before them, and each has the same tendency to enter.

If a Bernoulli trial is one traveler deciding whether to enter, the **Binomial distribution** counts the total number of travelers who walk in out of the group of 20. It aggregates multiple independent Bernoulli trials.



Theoretical Foundation

Type: Discrete probability distribution

Parameters: $n \in \{1, 2, 3, \dots\}$ (number of independent trials), $p \in [0, 1]$ (probability of success per trial)

Support: $k \in \{0, 1, 2, \dots, n\}$

Probability Mass Function (PMF):

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

where $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ accounts for every possible ordering of k successes across n trials.

Expectation (Mean): $E[X] = np$

Variance: $\text{Var}(X) = np(1 - p)$

Business Case Study: Quality Assurance in Electronics Manufacturing

Scenario: A hardware manufacturer tests sample batches of $n = 500$ smart sensors per production cycle. Historical sensor defect probability is $p = 0.015$.

Interpretation: The expected number of defective sensors per batch is $E[X] = 500 \times 0.015 = 7.5$. The plant manager checks the probability that a batch contains more than 15 defective units: $P(X > 15) = 1 - \sum_{k=0}^{15} P(X = k)$.

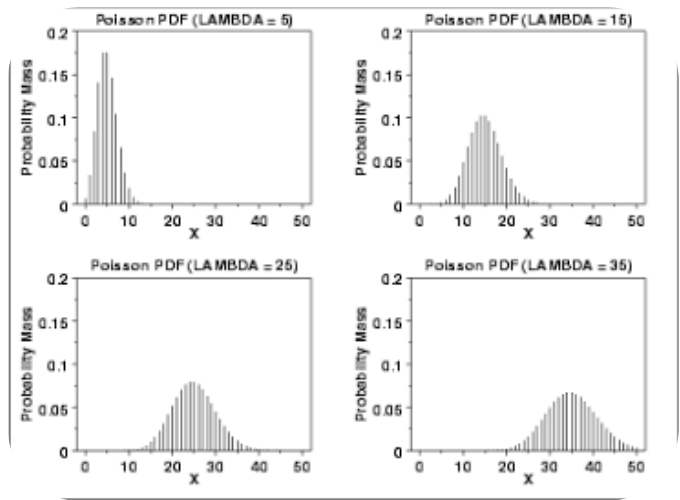
Benefit: It helps teams set clear acceptance sampling thresholds. Batches exceeding the rejection cutoff trigger immediate line inspections, protecting brand reputation while minimizing wasted scrap.

Module 3: The Poisson Distribution

Story & Conceptual Intuition

Step inside the coffee shop and watch the barista. People walk into the store continuously throughout the morning rush. You cannot easily count fixed "trials" because time flows continuously. However, you know from past experience that on average, 12 customers place an order every 30 minutes.

The **Poisson distribution** counts how many rare, discrete events happen within a fixed interval of continuous time, space, or volume, given a constant average rate λ .



Poisson PMFs at varying lambda values. Source: Information Technology Laboratory

Theoretical Foundation

- Type: Discrete probability distribution
- Parameter: $\lambda > 0$ (the expected rate of events per interval)
- Support: $k \in \{0, 1, 2, \dots\}$
- Probability Mass Function (PMF):

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

- Expectation (Mean): $E[X] = \lambda$
- Variance: $\text{Var}(X) = \lambda$

Because the mean equals the variance, checking whether $\bar{x} \approx s^2$ in real data is an easy way to see if a Poisson process fits.

Business Case Study: Cloud Server Traffic & API Capacity

- Scenario: A fintech payment gateway receives an average of $\lambda = 45$ authentication requests per second during peak hours.
- Interpretation: Incoming requests arrive independently at an average rate. Engineering uses the Poisson PMF to calculate the tail probability of receiving more than 70 requests in any given second: $P(X > 70)$.
- Benefit: It guides cloud infrastructure teams in configuring auto-scaling thresholds. Server clusters scale up just before traffic spikes cause request queuing, avoiding both server downtime and over-provisioning costs.

Module 4: The Continuous Uniform Distribution

Story & Conceptual Intuition

Imagine waiting at a bus stop where the bus arrives strictly every 15 minutes, but you lost your watch and arrive at a random moment. You might wait 0 seconds, 4.32 minutes, 11.89 minutes, or anywhere up to 15 minutes. Every specific interval of equal length has the exact same likelihood.

The **Continuous Uniform distribution** represents complete uncertainty within boundaries: every single point along the spectrum has an equal likelihood density.

Theoretical Foundation

Type: Continuous probability distribution

Parameters: $a, b \in \mathbb{R}$ with $a < b$ (minimum and maximum bounds)

Support: $x \in [a, b]$

Probability Density Function (PDF):

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

Cumulative Distribution Function (CDF):

$$F(x) = \frac{x - a}{b - a} \quad \text{for } x \in [a, b]$$

Expectation (Mean): $E[X] = \frac{a+b}{2}$

Variance: $\text{Var}(X) = \frac{(b-a)^2}{12}$

Business Case Study: Monte Carlo Project Scheduling

Scenario: A construction project manager estimates that pouring and setting a commercial foundation will take between $a = 14$ days and $b = 26$ days, with no historical reason to favor any specific duration in between.

Interpretation: The lead time is sampled uniformly from $\mathcal{U}(14, 26)$. The probability of completion within 20 days is:

$$P(X \leq 20) = \frac{20 - 14}{26 - 14} = \frac{6}{12} = 50\%$$

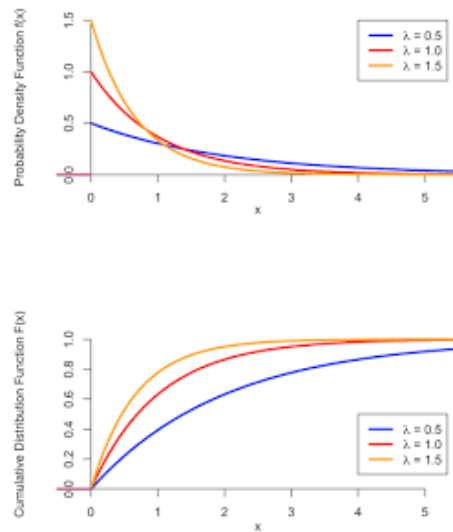
Benefit: Uniform distributions serve as non-informative priors in risk modeling and Monte Carlo simulations. They prevent planners from assuming false precision when data is limited.

Module 5: The Exponential Distribution

Story & Conceptual Intuition

Return to our barista for a moment. While the Poisson distribution counts *how many* customers walk in per hour, the **Exponential distribution** measures *how long* you have to wait between customer arrivals.

It carries an interesting property known as **memorylessness**: if you have already waited 10 minutes without a customer walking in, the probability that someone arrives in the next 2 minutes remains identical to the first 2 minutes you waited. The process does not "remember" how long it has been waiting.



Exponential distribution PDF and CDF curves. Source: dlsun.github.io

Theoretical Foundation

Type: Continuous probability distribution

Parameter: $\lambda > 0$ (rate parameter), or scale $\beta = \frac{1}{\lambda}$ (mean wait time)

Support: $x \in [0, \infty)$

Probability Density Function (PDF):

$$f(x) = \lambda e^{-\lambda x} \quad \text{for } x \geq 0$$

Cumulative Distribution Function (CDF):

$$F(x) = 1 - e^{-\lambda x}$$

Memoryless Property: $P(X > s + t \mid X > s) = P(X > t)$

Expectation (Mean): $E[X] = \frac{1}{\lambda}$

Variance: $\text{Var}(X) = \frac{1}{\lambda^2}$

Business Case Study: Customer Service Call Queues

Scenario: An airline service center handles incoming calls where the average duration of a support ticket is 6 minutes, meaning $\lambda = \frac{1}{6} \approx 0.167$ calls completed per minute.

Interpretation: The time T until an agent wraps up an active call follows an Exponential distribution. The probability that an interaction wraps up within 3 minutes is:

$$P(T \leq 3) = 1 - e^{-0.167 \times 3} = 1 - e^{-0.5} \approx 39.3\%$$

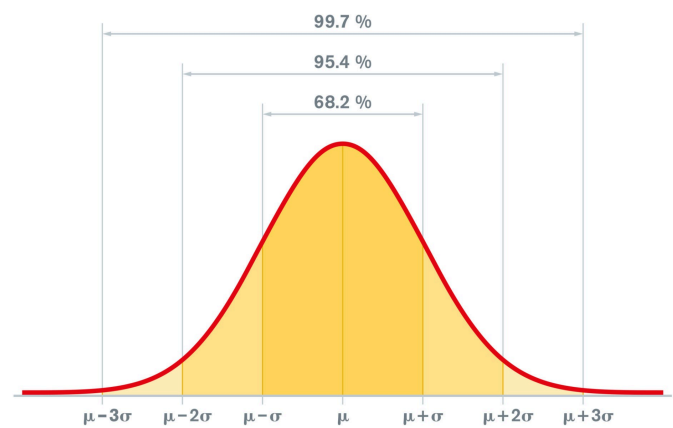
Benefit: Operations leaders use this distribution to optimize customer wait times via queueing theory ($M/M/k$ models). It helps allocate the right number of floor staff to keep call abandons under target service level agreements.

Module 6: The Normal (Gaussian) Distribution

Story & Conceptual Intuition

Imagine a pastry chef baking loaves of artisanal sourdough bread. The recipe calls for 500 grams of flour. Tiny fluctuations happen on every bake: a breeze shifts the scale balance by 0.2 grams, humidity alters water absorption, hand kneading stretches the dough slightly differently, and ambient oven temperatures fluctuate.

None of these minor disturbances dominate. When many small, independent additive forces push and pull simultaneously, the outcomes gather into a smooth, symmetrical bell curve centered on the target. This behavior is captured by the **Central Limit Theorem**.



Normal distribution 68-95-99.7 empirical rule. Source: PeterHermesFurian / Getty Images

Theoretical Foundation

Type: Continuous probability distribution

Parameters: $\mu \in \mathbb{R}$ (mean/center), $\sigma > 0$ (standard deviation/spread)

Support: $x \in (-\infty, \infty)$

Probability Density Function (PDF):

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x - \mu}{\sigma}\right)^2\right)$$

Empirical Rule:

$\approx 68.27\%$ of values fall within $\mu \pm 1\sigma$

$\approx 95.45\%$ of values fall within $\mu \pm 2\sigma$

$\approx 99.73\%$ of values fall within $\mu \pm 3\sigma$

Expectation (Mean): $E[X] = \mu$

Variance: $\text{Var}(X) = \sigma^2$

Business Case Study: Six Sigma Process Control

Scenario: An automated bottling facility fills 2-liter soda bottles. The machine runs with $\mu = 2005$ mL and $\sigma = 2.5$ mL. The legal lower specification limit (LSL) is 1995 mL.

Interpretation: The fill volume $X \sim \mathcal{N}(2005, 2.5^2)$. To assess risk, compute the Z-score:

$$Z = \frac{1995 - 2005}{2.5} = -4.0$$

The probability of an underfilled bottle falling past -4σ is roughly 0.00317%, or about 32 per million bottles.

Benefit: By maintaining high statistical capability (C_p/C_{pk} metrics), the business avoids consumer protection fines for underfilled containers while reducing overflow product waste.

Module 7: The Log-Normal Distribution

Story & Conceptual Intuition

Let us return once more to wealth, stock prices, or real estate values. Why do they rarely fit a balanced, symmetrical bell curve?

In finance and biology, growth is **multiplicative**, not additive. A 10% gain on \$1,000,000 is \$100,000, while a 10% gain on \$10,000 is only \$1,000. Because variables cannot drop below zero and growth compounds over time, the data develops a long right tail.

If taking the natural logarithm of a skewed variable transforms it into a standard bell curve, that variable follows a **Log-Normal distribution**.

Theoretical Foundation

Type: Continuous probability distribution

Definition: If $Y \sim \mathcal{N}(\mu, \sigma^2)$, then $X = e^Y$ is Log-Normally distributed: $X \sim \text{Lognormal}(\mu, \sigma^2)$.

Support: $x \in (0, \infty)$

Probability Density Function (PDF):

$$f(x) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln x - \mu)^2}{2\sigma^2}\right) \quad \text{for } x > 0$$

Key Relationship: Mode < Median (e^μ) < Mean

Expectation (Mean): $E[X] = \exp\left(\mu + \frac{\sigma^2}{2}\right)$

Variance: $\text{Var}(X) = (e^{\sigma^2} - 1) \exp(2\mu + \sigma^2)$

Business Case Study: Real Estate Valuation & Insurance Claims

Scenario: An insurance group models commercial property flood damage claims. Most payouts are small (repairs under \$15,000), but occasional hurricanes cause claims exceeding \$12,000,000.

Interpretation: The raw claim amounts are heavily right-skewed and bounded strictly at zero. Taking the natural log of historic claim values yields an approximately symmetric normal distribution with mean log-claim $\mu = 9.8$ and log-variance $\sigma = 1.1$.

Benefit: Applying a Log-Normal model prevents the underestimation of extreme catastrophe risk. The firm can price insurance premiums and size reserve capital accurately, preventing insolvency during severe loss years.

Core Distribution Comparison

Distribution	Type	Support	Key Parameters
Bernoulli	Discrete	$\{0, 1\}$	p (success probability)
Binomial	Discrete	$\{0, 1, \dots, n\}$	n (trials), p (success probability)
Poisson	Discrete	$\{0, 1, 2, \dots\}$	λ (rate per interval)
Uniform	Continuous	$[a, b]$	a (min), b (max)
Exponential	Continuous	$[0, \infty)$	λ (rate) or β (scale)
Normal	Continuous	$(-\infty, \infty)$	μ (mean), σ (standard deviation)
Log-Normal	Continuous	$(0, \infty)$	μ (scale), σ (shape)